MADE BY :

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Project Brief Description:

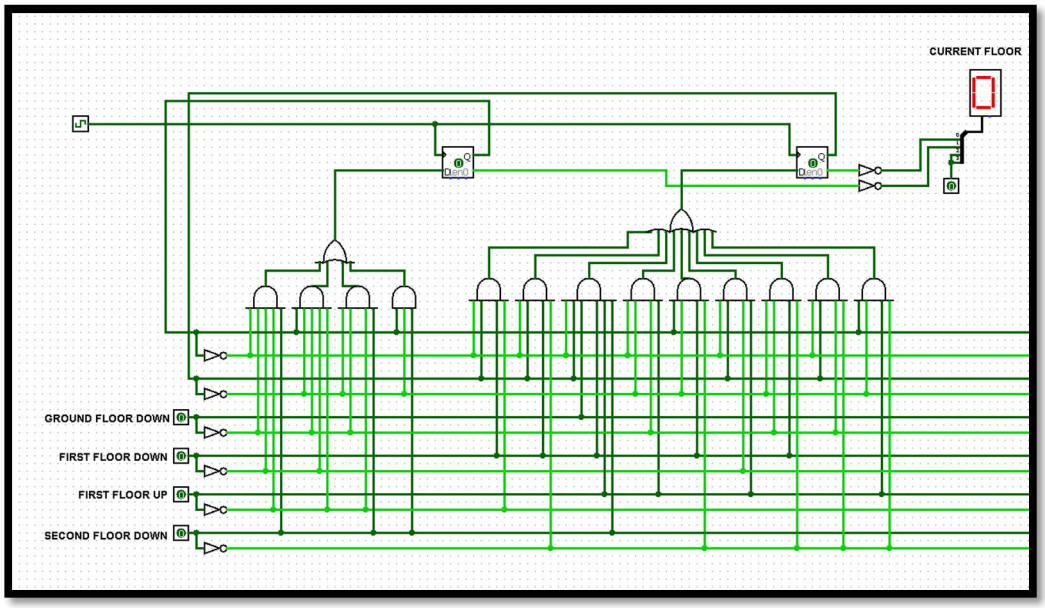
The project **"Elevator Control System"** is a digital circuit designed using **Logisim** software to simulate the logical operation of an elevator moving between multiple floors.

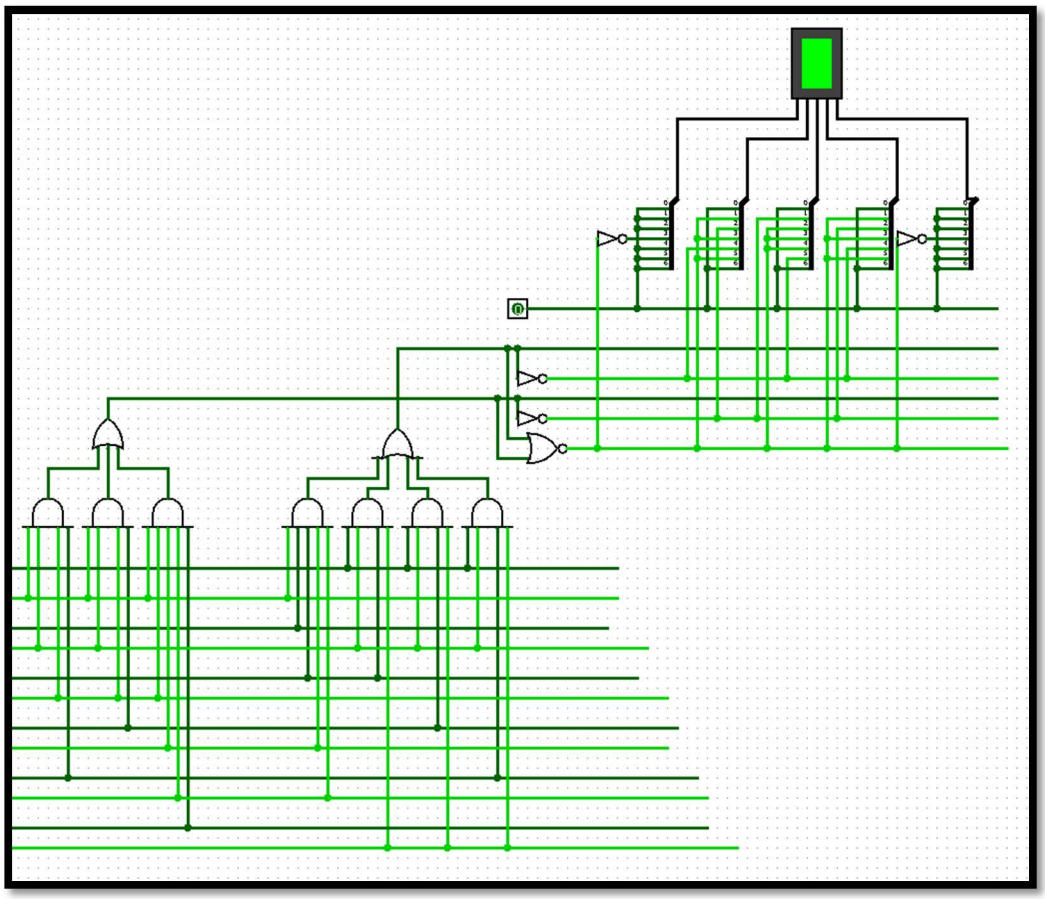
The system uses **combinational and sequential logic** to control the elevator's movement, direction, and floor indication.

It aims to demonstrate the practical application of **digital electronics** concepts like **logic gates, flip-flops**, and with the help of **sequential circuit** to build a real-world control mechanism.

This project provides a logical understanding of how elevators decide their direction, respond to floor calls, and ensure smooth movement between floors using digital signals.

Circuit Screenshot:





Working Explanation:

Overview:

The circuit simulates an elevator that operates between **Ground Floor, First Floor, and Second Floor**.

Each floor has **Up and Down call buttons**, and the elevator responds based on the current position and requested direction.

Step-by-Step Working:

1. **Input Section:**
    - Each floor has input buttons such as:
        - **Ground Floor UP**
        - **First Floor UP / DOWN**
        - **Second Floor DOWN**
    - When a floor button is pressed, it sends a HIGH logic signal to request elevator movement.
2. **Control Logic:**
    - The control logic uses **AND, OR, and NOT gates** to determine the direction and the next floor.

Next state equation:

$$\boxed{F(A) = A'BDE' + A'BDF' + A'BCF + A'B'C'E + AB'EF' + AB'DF' + A'BD'E + B'C'DF' + A'BC'F'}$$

$$\boxed{F(B) = A'BCE + AB'DF' + AB'EF' + A'BDE' + A'BD'E + A'BC'F' + A'C'DF' + A'B'C'E}$$

	F(A)	F(B)
GROUND FLOOR	0	0
FIRST FLOOR	0	1
THIRD FLOOR	1	0

- Based on the request and current floor, the elevator moves **upward or downward**.

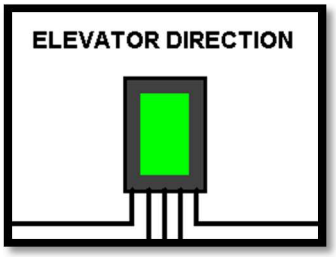
3. **Direction Indicator:**

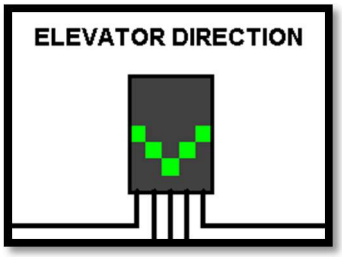
   - The **ELEVATOR DIRECTION** block (green LED) indicates whether the elevator is moving **up** or **down**.
   - Logic gates connected to the floor selection inputs control this signal.

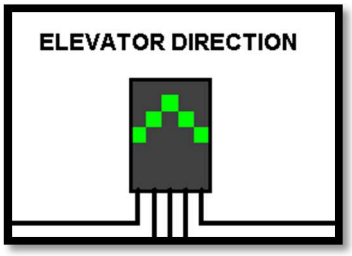
$$\boxed{F(first) = A'C'D'E'F + A'B'C'E + A'B'C'D}$$

$$\boxed{F(A,B,C,D,E,F) = A'BCD'E' + AB'CF' + AB'DF' + AB'EF'}$$

	F(first bit)	F(second bit)
NO change	0	0
Down Direction	0	1
Up Direction	1	0

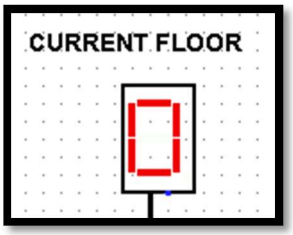


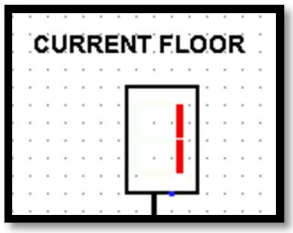


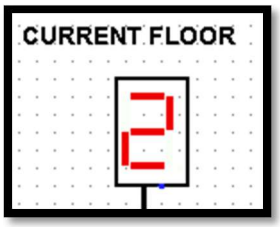


4. **Current Floor Display:**

   - The **CURRENT FLOOR** indicator (red LED or 7-segment display) shows the present floor of the elevator.
   - This is achieved by connecting **output logic** to represent floor status.







5. **Sequential Movement:**

- When multiple floors are requested, the elevator logic prioritizes floors in the current direction (up or down) before reversing direction.
- This mimics the working of a real elevator control system.

Components Used:

Component Name	Symbol/Type	Function / Purpose
Logic Gates (AND, OR, NOT)	Combinational Devices	Used to implement decision-making logic for direction and floor movement
Flip-Flops (D)	Sequential Elements	Store the current floor status
LEDs Matrix	Output Indicators	Display elevator direction and floor position
Switches	Input Devices	Represent floor call buttons (Up/Down)
Clock Pulse	Input Timing Source	Synchronizes sequential operations
Wires / Connectors	—	Used for interconnections between components

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Conclusion:

The **Elevator Control System** project effectively demonstrates how digital electronics can be used to design control logic for automated systems.

By combining **combinational and sequential circuits**, the elevator can make logical decisions similar to real-world implementations.

This project enhances the understanding of circuit design, logical reasoning, and real-time problem-solving in digital systems.